Distance & similarity measures – Primer

Concepts with python code demos

Topics

- Euclidean distance
- Taxicab / Manhattan
- Minkowski
- Jaccard index
- Hamming distance
- Levenshtein Distance (Edit Distance)
- Canberra Distance
- Chebyshev Distance (Infinity Norm)

Many other types of similarity measures exists for text and images and some complex data distributions

TO BE COVERED IN THE RELEVANT COURSES

Pre-reqs

Python, numpy, pandas, plots

Basic stats for DS

Elementary idea of Machine learning

Euclidean distance

a measure of the straight-line distance between two data points (rows, samples etc)

Demo using python/sklearn

(Basics of co-ordinate geometry)

Define

Euclidean distance is a measure of straight-line distance between two points in Euclidean space.

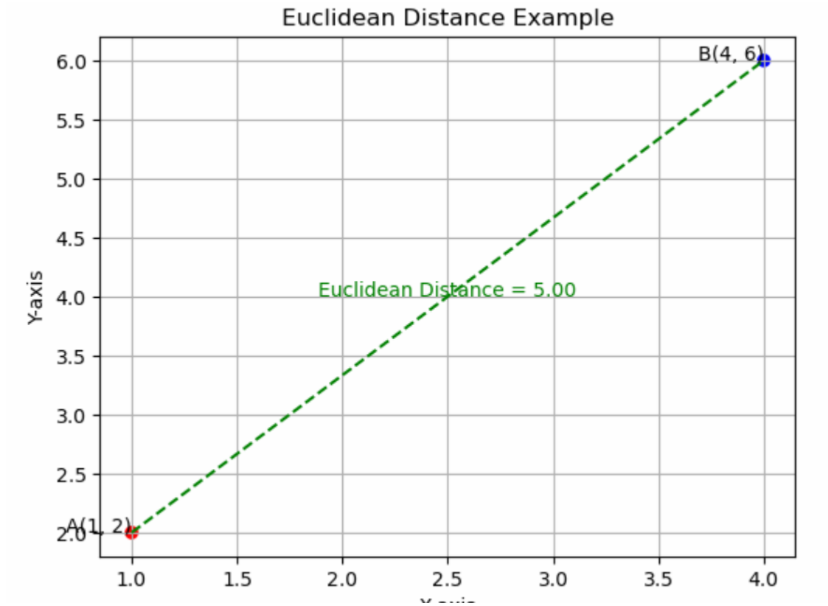
It is the most common distance metric and is widely used in various applications, including clustering, classification, and dimensionality reduction.

Formula

For two data points/ rows/ samples/documents …

A(*x*1, *y*1) and *B*(*x*2, *y*2) in a 2D space, the Euclidean distance is calculated using the formula:

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Let's consider two points in a 2D space: *A*(1, 2) and *B*(4, 6)

$$d(A, B) = \sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

Strengths of Euclidean Distance

Intuitive Geometric Interpretation:

- Euclidean distance corresponds to the straight-line distance between two points in Euclidean space, which is a familiar geometric concept.

Simple and Computationally Efficient:

- The calculation of Euclidean distance is straightforward and computationally efficient. It involves basic arithmetic operations (subtraction, squaring, and square root),

Applicability to Various Data Types:

- Euclidean distance can be applied to a wide range of data types, including numerical, categorical (with appropriate encoding), and spatial data.

Well-Suited for Clustering:

- Euclidean distance is commonly used in clustering algorithms, such as k-means, where it helps define the concept of compact and well-separated clusters.

Relationship to Similarity Measures:

- In many cases, a high similarity corresponds to a low Euclidean distance and vice versa.

Widely Adopted and Understood:

- Euclidean distance is a well-established and widely adopted metric in various fields, including mathematics, physics, and computer science.

weaknesses of Euclidean distance

Features with larger scales may dominate the distance calculation, leading to biased results.

Euclidean distance assumes a linear relationship between features, which may not be appropriate for all types of data.

Outliers can distort the results and affect the performance of algorithms that rely on Euclidean distance, especially in applications like clustering.

Euclidean distance is designed for numerical data and may not be directly applicable to categorical data. While encoding schemes (e.g., one-hot encoding) can be used to represent categorical variables numerically, the resulting distances may not always reflect meaningful dissimilarities.

In high-dimensional spaces, the Euclidean distance between points tends to increase, leading to a phenomenon known as the "curse of dimensionality." As the number of features grows, the meaningfulness of Euclidean distance diminishes.

Euclidean distance assumes metric properties, such as the triangle inequality. In non-metric spaces or spaces with non-Euclidean geometry, these assumptions may not hold.

Effect of Scale Differences on Euclidean Distance

- Euclidean distance is sensitive to the scale of features, meaning that the magnitudes of individual features can influence the overall distance between points.

- **Example:** Consider two points in a 2D space:
  - Point A: (2, 3)
  - Point B: (6, 8)

$$
\begin{aligned}
d_{AB} &= \sqrt{(6-2)^2 + (8-3)^2} \\
&= \sqrt{4^2 + 5^2} \\
&= \sqrt{16 + 25} \\
&= \sqrt{41} \approx 6.4
\end{aligned}
$$

scale the second feature

- (y-coordinate) by a factor of 10
  - Point A: (2, 3)
  - Point B: (6, 80)

- The new Euclidean distance (d'_{AB}) is calculated as:

$$
\begin{aligned}
d'_{AB} &= \sqrt{(6-2)^2 + (80-3)^2} \\
&= \sqrt{4^2 + 77^2} \\
&= \sqrt{16 + 5929} \\
&= \sqrt{5945} \approx 77.1
\end{aligned}
$$

Euclidean distance and non-linear data

In a linear relationship, the data points tend to lie along or close to a straight line.

Euclidean distance effectively measures the distance between points in such scenarios.

Example:

- Suppose you have data points representing the relationship between hours of study (X-axis) and exam scores (Y-axis).
- If the relationship is roughly linear, Euclidean distance can accurately measure the distance between two points in this space.

Challenges in Non-Linear Relationships

In cases where the relationships between variables are non-linear, Euclidean distance may not fully capture the underlying patterns.

Non-linear relationships may involve curves, bends, or other complex shapes

Example:

- Consider a dataset with points forming a circle in a 2D space.
- The Euclidean distance between points on opposite sides of the circle may not accurately represent their similarity, as it measures the straight-line distance through the circle rather than following the circular path.

Outliers and Euclidean distance

- Outliers can have a disproportionate impact on the calculation of Euclidean distance because the distance is based on the squared differences between corresponding coordinates.
- **Example:** Consider a 2D dataset with points:
    - A: (1, 2)
    - B: (3, 4)
    - C: (15, 20) (Outlier)
- Euclidean distance between points A and B is smaller compared to the distance between points A and C.
- The outlier C, due to its extreme values, significantly affects the overall distance calculation.

$$\text{Euclidean Distance}(A, B) = \sqrt{(3-1)^2 + (4-2)^2} = \sqrt{2^2 + 2^2} = \sqrt{8}$$

$$\text{Euclidean Distance}(A, C) = \sqrt{(15-1)^2 + (20-2)^2} = \sqrt{14^2 + 18^2} = \sqrt{676 + 324} = \sqrt{1000} = 31.62$$

Euclidean distances – categorical variables

- Suppose we have a dataset representing the preferences of individuals for different animals.
- The features include "Favorite Pet" (categorical) and "Number of Pets" (numerical).

Person	Favorite Pet	Number of Pets
Alice	Dog	2
Bob	Cat	1
Carol	Fish	3
Dave	Dog	0

Encoding Categorical Data

- To use Euclidean distance, we need to encode the categorical variable "Favorite Pet."
- One common method is one-hot encoding:

Person	Dog	Cat	Fish	Number of Pets
Alice	1	0	0	2
Bob	0	1	0	1
Carol	0	0	1	3
Dave	1	0	0	0

Euclidean Distance Calculation

X, *Y*, and *Z* represent the one-hot encoded features for "Dog," "Cat," and "Fish" respectively.

$$\text{Euclidean Distance}(A, B) = \sqrt{(X_A - X_B)^2 + (Y_A - Y_B)^2 + (Z_A - Z_B)^2}$$

Example

$$\text{Euclidean Distance}(Alice, Bob) = \sqrt{(1-0)^2 + (0-1)^2 + (0-0)^2 + (2-1)^2}$$
$$= \sqrt{1+1+1} = \sqrt{3} \approx 1.732$$

Issue

While the Euclidean distance calculation is technically possible, the resulting distances may not reflect meaningful dissimilarities between individuals.

The encoding suggests equal dissimilarity between "Dog" and "Fish" as between "Dog" and "Cat," which may not align with the actual preferences of individuals.

Impact on Euclidean Distance in High Dimensions

- Consider a set of points in a two-dimensional space (2D), and then extend the analysis to a higher-dimensional space (e.g., 10D or 100D).
- The impact on Euclidean distance becomes more apparent as the number of dimensions increases.
- Let's start with a simple 2D example where we have two points A and B with coordinates:
  - Point A: (1, 2)
  - Point B: (4, 6)

$$
\begin{aligned}
d_{2D}(A, B) &= \sqrt{(4-1)^2 + (6-2)^2} \\
&= \sqrt{3^2 + 4^2} \\
&= \sqrt{9 + 16} \\
&= \sqrt{25} \\
&= 5
\end{aligned}
$$

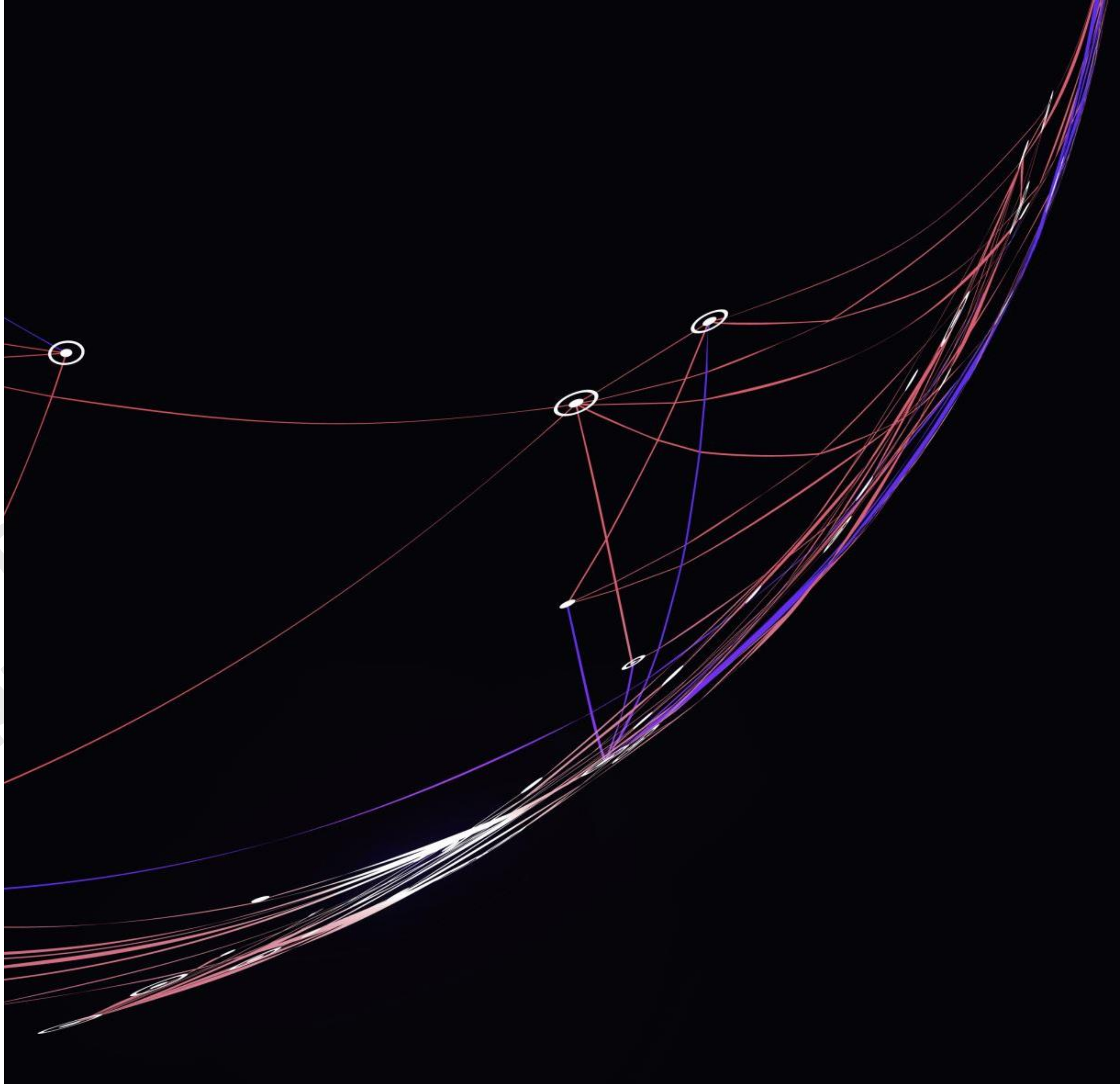
High-Dimensional Space (10D)

- Now, extend the example to a 10-dimensional space, where points A and B have coordinates in each dimension:
  - Point A: (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
  - Point B: (4, 6, 8, 10, 12, 14, 16, 18, 20, 22)
- **Magnitude of Differences Matters:**
  - In higher dimensions, the magnitude of differences (*Bi*−*Ai*) becomes crucial.
  - Even small differences along each dimension contribute to a significant increase in the overall distance.

$$
\begin{aligned}
d_{10\mathrm{D}}(A, B) &= \sqrt{\textstyle\sum_{i=1}^{10} (B_i - A_i)^2} \\
&= \sqrt{(4-1)^2 + (6-2)^2 + \ldots + (22-10)^2} \\
&= \sqrt{3^2 + 4^2 + \ldots + 12^2} \\
&= \sqrt{9 + 16 + \ldots + 144}
\end{aligned}
$$

Equidistant Points

- As the number of dimensions increases, the Euclidean distance tends to make <u>all points approximately equidistant</u> from each other.
- This is because the contribution of each dimension to the distance becomes more balanced, leading to a <u>flattening effect</u> in the distance landscape.



Sparse data and Euclidean distance

Sparse data refers to datasets where a significant portion of the feature values are zero or near-zero.

often occurs in high-dimensional datasets, such as text data, where each document is represented by a vector of word frequencies.

Document	apple	banana	orange	juice	drink	fruit
Document 1	1	1	0	0	0	0
Document 2	0	0	1	1	0	0
Document 3	0	0	0	1	1	1

Example

- Consider a small collection of documents with a vocabulary of six words: "apple," "banana," "orange," "juice," "drink," and "fruit."
- The documents are:
  - Document 1: "I like apple and banana.“
  - Document 2: "Orange juice is tasty.“
  - Document 3: "Drink fruit juice every day."

Result

matrix is sparse because most entries are zero.

Each row corresponds to a document.

Each column corresponds to a unique word in the vocabulary.

The values represent the frequency of each word in the respective document.

using sparse data, considerations of alternative distance metrics (e.g., cosine similarity) become important for meaningful analysis.

Demo using python/sklearn

(Euclidean distance, Scale diff, curse of dimensionality, outliers, sparse)

Taxicab / Manhattan distance

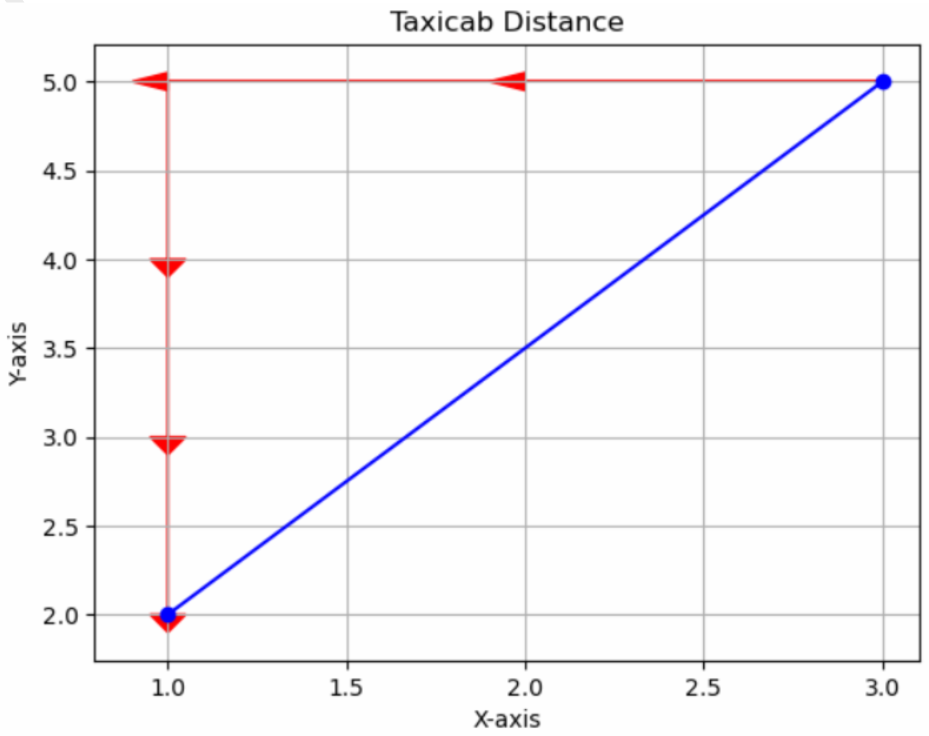
The name "Manhattan distance" comes from the grid layout of the streets in Manhattan.

Define

Taxicab distance, also known as Manhattan distance or L1 distance, is a measure of the distance between two points in a grid-based system (like a city grid) measured along the grid lines.

Mathematically, the Taxicab distance between two points $(x1,y1)$ and $(x2,y2)$ in a 2D space is given by:

$$\text{Taxicab distance} = |x1-x2| + |y1-y2|$$



type of data Manhattan distance is suitable

Grid-Based or Categorical Data:

- well-suited for grid-based systems, such as city layouts, game boards, or any situation where movement is restricted to horizontal and vertical steps.
- suitable for categorical data

Sparse Data:

- Manhattan distance can be more effective than Euclidean distance.
- As Manhattan distance only considers the non-zero dimensions, making it less sensitive to the presence of zeros.

Robustness to Outliers:

- Manhattan distance is less sensitive to outliers compared to Euclidean distance.

Feature Importance is Equal:

- When all dimensions (features) are considered equally important, Manhattan distance may be appropriate.

Minkowski

a generalized distance metric

Define

The Minkowski distance is a generalized distance metric that encompasses both the Manhattan (L1 norm) and Euclidean (L2 norm) distances as special cases.

distance between two points $P=(p_1,p_2,\ldots,p_n)$ and $Q=(q_1,q_2,\ldots,q_n)$ in an n-dimensional space is given by:

$$D = \left(\sum_{i=1}^{n} |p_i - q_i|^r\right)^{\frac{1}{r}}$$

Properties of Minkowski Distance:

1. **Manhattan Distance ($r = 1$):**
   - $D_{\text{Manhattan}}(P, Q) = \sum_{i=1}^{n} |p_i - q_i|$
2. **Euclidean Distance ($r = 2$):**
   - $D_{\text{Euclidean}}(P, Q) = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}$
3. **Chebyshev Distance ($r = \infty$):**
   - $D_{\text{Chebyshev}}(P, Q) = \max_i |p_i - q_i|$
4. **General Case ($r > 0$):**
   - For any positive r, the Minkowski distance generalizes the Manhattan and Euclidean distances.

Demo using python/sklearn

(Minkowski)

Jaccard index

- also known as the Jaccard similarity coefficient, is a measure of similarity between two sets.
- It is defined as the size of the intersection of the sets divided by the size of the union of the sets.

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

- **Examples:**
- Let's consider two sets, *A*={1,2,3,4} and *B*={3,4,5,6}.

1. **Intersection:**
   - $A \cap B = \{3, 4\}$
2. **Union:**
   - $A \cup B = \{1, 2, 3,$
3. **Jaccard Index:**
   - $J(A,B) = \frac{|\{3,}{|\{1,2,3,}$

situations where the Jaccard index is used

- In document analysis, the Jaccard index can measure the similarity of sets of words or terms used in different documents.

- **Document 1:** {apple, orange, banana, pear}
- **Document 2:** {orange, banana, grape, kiwi}

- **Calculation:**
  - Intersection: {orange, banana}
  - Union: {apple, orange, banana, pear, grape, kiwi}
  - Jaccard Index: 2/6=1/3

Recommendation Systems

- the Jaccard index can measure the similarity of item sets between users.

- Example
  - **User 1's Rated Movies:** {Star Wars, Avatar, Inception}
  - **User 2's Rated Movies:** {Inception, The Matrix, Jurassic Park}

- **Calculation:**
  - Intersection: {Inception}
  - Union: {Star Wars, Avatar, Inception, The Matrix, Jurassic Park}
  - Jaccard Index: 1/

Biology and Bioinformatics

- In genomics and bioinformatics, the Jaccard index is employed to measure the similarity of gene sets or biological pathways.
- useful for comparing experimental results, identifying common biological functions, and clustering genes.
- **Example**
  - **Gene Set 1:** {geneA, geneB, geneC, geneD}
  - **Gene Set 2:** {geneC, geneD, geneE, geneF}
- **Calculation:**
  - Intersection: {geneC, geneD}
  - Union: {geneA, geneB, geneC, geneD, geneE, geneF}
  - Jaccard Index: 2/6

Types of data where Jaccard index is suitable

1. **Text Data:**
    1. Document analysis, natural language processing, and text mining tasks
2. **Recommendation Systems:**
    1. User-item interactions, such as movie preferences, product ratings, or liked items,
3. **Genomic Data:**
    1. In bioinformatics, the Jaccard index is employed to compare sets of genes, biological pathways, or genomic features.
4. **Social Network Analysis:**
    1. Assessing the similarity between sets of connections or friends in a social network is a common use case.

Hamming distance

measure the difference between two strings

Define

a metric used to measure the difference between two strings of equal length.

It counts the number of positions at which the corresponding symbols (characters or bits) are different.

applicable to binary strings, DNA sequences, and even strings of characters, as long as they have the same length.

Calculation

$$H(X,Y) = \sum_{i=1}^{n} \delta(x_i, y_i)$$

xi and *yi* are the symbols at position *i* in strings *X* and *Y*, respectively, and

δ(*xi*,*yi*) function, equals 0 if *xi*=*yi* and 1 if *they are not same*.

Example

Let's consider two binary strings:

String X: 11001

String Y: 10101

Calculation: $H(X,Y) = \delta(1,1) + \delta(1,0) + \delta(0,1) + \delta(0,0) + \delta(1,1) = 2$

In this example, the Hamming distance between the two binary strings is 2, indicating that there are two positions where the corresponding symbols differ.

Properties of Hamming Distance

01

Hamming distance is always a non-negative integer.

02

It is defined for strings of equal length.

03

The larger the Hamming distance, the more dissimilar the strings.

Canberra distance

$$D(x,y) = \sum_{i=1}^{n} \frac{|x_i - y_i|}{|x_i| + |y_i|}$$

Canberra distance is a metric used to measure the dissimilarity between two vectors with numerical values.

It is particularly suitable for datasets where the variables have <u>different scales</u> and is often used in data analysis, clustering, and similarity measurement.

Example

$$x = [2, 4, 6]$$
$$y = [1, 3, 7]$$

$$D(x, y) = \frac{|2-1|}{|2|+|1|} + \frac{|4-3|}{|4|+|3|} + \frac{|6-7|}{|6|+|7|}$$

$$D(x, y) = \frac{1}{3} + \frac{1}{7} + \frac{1}{13}$$

Now, you sum up these terms:

$$D(x, y) = \frac{1}{3} + \frac{1}{7} + \frac{1}{13} \approx 0.424$$

So, the Canberra distance between x and y is approximately 0.424.

Chebyshev Distance (Infinity Norm)

THE CHEBYSHEV DISTANCE, ALSO KNOWN AS THE INFINITY NORM OR L∞ NORM, IS A DISTANCE METRIC THAT MEASURES THE MAXIMUM ABSOLUTE DIFFERENCE BETWEEN CORRESPONDING ELEMENTS OF TWO VECTORS.

IT IS NAMED AFTER THE MATHEMATICIAN CHEBYSHEV.

$$D_\infty(X, Y) = \max_{i=1}^{n} |x_i - y_i|$$

$$D_\infty(X,Y) = \max_{i=1}^{n} |x_i - y_i|$$

Example Calculation

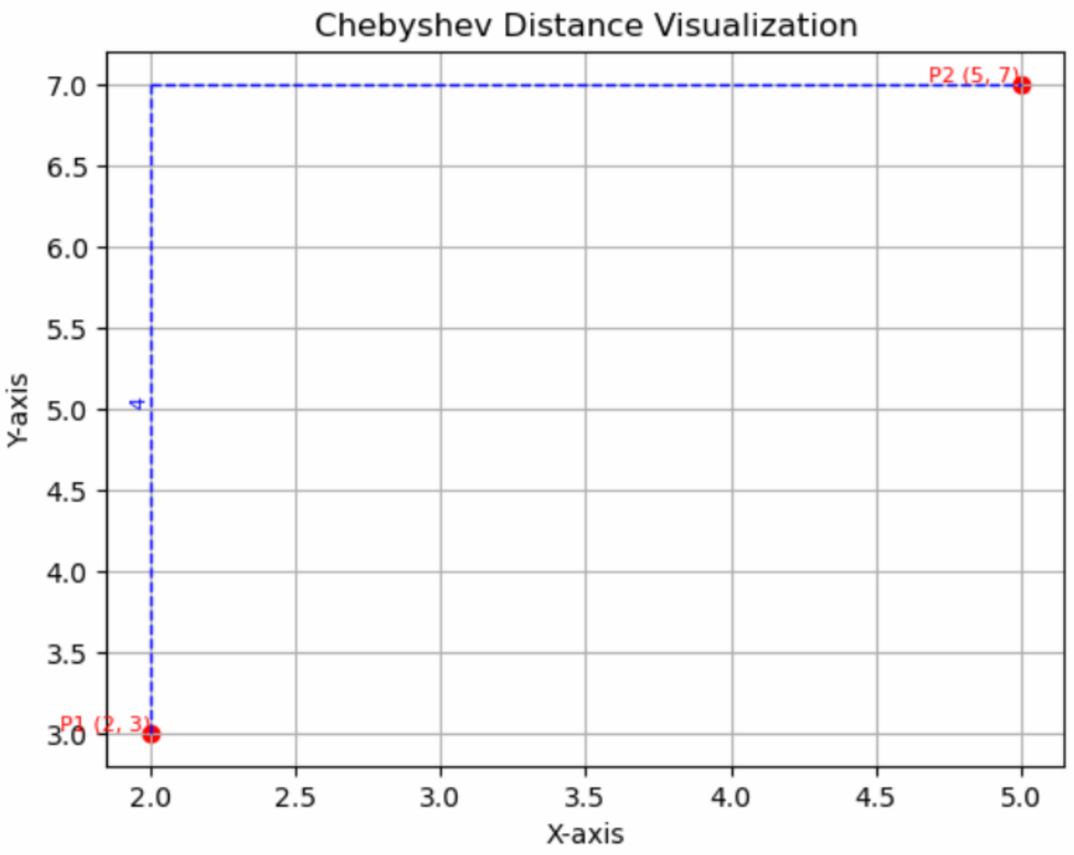
- Let's consider two vectors:
  - X=[4,7,2]
  - Y=[1,5,8]

Calculation:

1. **Absolute Differences:**
   - $|x_1 - y_1| = |4-1| = 3$
   - $|x_2 - y_2| = |7-5| = 2$
   - $|x_3 - y_3| = |2-8| = 6$
2. **Maximum Absolute Difference:**
   - $D_\infty(X,Y) = \max(3,2,6) = 6$

Result:

The Chebyshev distance between vectors X and Y is 6.



When to use Chebyshev Distance (Infinity Norm)

Outlier Detection:

is effective in identifying outliers or anomalies in a dataset. By measuring the maximum difference along any dimension, it provides a robust measure that is less sensitive to extreme values.

Image Processing:

When comparing images, especially in scenarios where pixel intensity values are crucial, Chebyshev Distance can be employed to measure dissimilarity.

Signal Processing:

in signal processing when comparing signals that may experience variations in amplitude along their timelines. It helps capture the worst-case scenario of differences in signal magnitudes.

Demo using python/sklearn

(Chebyshev distance)

Thanks !!